

Performance Modeling of Computer Systems (PMCS)

Complete Study Guide

All Lectures: January 2 – February 24, 2026

Topics Covered:

Queueing Theory • Stochastic Processes • Poisson Processes
Markov Chains (DTMC & CTMC) • Stationary Distributions
Renewal Theory • Little's Law • M/M/1 & M/M/ ∞ Queues
Tandem Queues • Stopping Times • Reinforcement Learning

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Introduction to Queueing Theory (Lecture: Jan 2)

What is Queueing Theory?

Queueing Theory is the mathematical study of *waiting lines* (queues). It models systems where “customers” arrive, possibly wait, receive “service”, and then depart.

Real-world examples:

- Packets waiting in a network router buffer
- HTTP requests waiting at a web server
- Processes waiting for CPU time in an operating system
- Customers in a bank queue

Why Do We Need Queueing Theory?

When you design a computer system (server, network, data center), you need to answer questions like:

- How long will a user wait on average?
- How many servers do I need so that 99% of requests are served within 1 second?
- What happens if the arrival rate doubles?

Queueing theory gives us **mathematical formulas** to answer these questions *without* building the actual system.

Basic Components of a Queueing System

Component	Description
Arrival Process	How customers arrive (e.g., Poisson process with rate λ)
Service Process	How long service takes (e.g., Exponential with rate μ)
Number of Servers	How many servers (1, k , or ∞)
Queue Capacity	Maximum number of customers that can wait (finite or ∞)
Queue Discipline	Order of service: FCFS, LCFS, Priority, etc.

Kendall's Notation

Kendall's Notation: A/B/c/K/N/D

A shorthand to describe a queue:

$$\underbrace{A}_{\text{Arrival}} / \underbrace{B}_{\text{Service}} / \underbrace{c}_{\text{Servers}} / \underbrace{K}_{\text{Capacity}} / \underbrace{N}_{\text{Population}} / \underbrace{D}_{\text{Discipline}}$$

Common symbols:

- **M** = Markovian (Memoryless) $\Rightarrow$ Exponential distribution
- **D** = Deterministic (constant times)
- **G** = General distribution

Example: M/M/1 = Poisson arrivals, Exponential service, 1 server, infinite capacity, FCFS.

Key Performance Metrics

Performance Metrics

Symbol	Name	Meaning
λ	Arrival rate	Average number of customers arriving per unit time
μ	Service rate	Average number of customers a server can serve per unit time
ρ	Utilization	Fraction of time the server is busy: $\rho = \lambda/\mu$
L	Avg. number in system	Average number of customers (waiting + being served)
L_q	Avg. number in queue	Average number of customers waiting only
W	Avg. time in system	Average time a customer spends (waiting + service)
W_q	Avg. waiting time	Average time a customer waits in queue only
P_b	Blocking probability	Probability a customer is rejected (finite capacity)

Stability Condition

A queue is **stable** (does not grow to infinity) if and only if:

$$\rho = \frac{\lambda}{\mu} < 1$$

If $\rho \geq 1$, the queue grows without bound — the server cannot keep up with arrivals.

Stochastic Processes (Lecture: Jan 6)

Stochastic Process

A **stochastic process** is a collection of random variables $\{X(t) : t \in T\}$ indexed by a parameter t (usually time).

Think of it as a “*random function of time*” — at each time t , the value $X(t)$ is random.

- **Discrete-time:** $T = \{0, 1, 2, \dots\}$ — e.g., daily stock prices
- **Continuous-time:** $T = [0, \infty)$ — e.g., number of packets at a router at time t
- **Discrete state space:** $X(t) \in \{0, 1, 2, \dots\}$ — countable set of values
- **Continuous state space:** $X(t) \in \mathbb{R}$ — real-valued

Independent Increments

Independent Increments

A process $\{X(t)\}$ has **independent increments** if for any times $t_1 < t_2 < \dots < t_n$, the increments:

$$X(t_2) - X(t_1), \quad X(t_3) - X(t_2), \quad \dots, \quad X(t_n) - X(t_{n-1})$$

are *mutually independent* random variables.

Intuition: What happens in one time interval does not affect what happens in another disjoint interval.

Stationary Increments

Stationary Increments

A process has **stationary increments** if the distribution of $X(t+s) - X(t)$ depends *only on s* (the length of the interval), not on t (when the interval starts).

Intuition: The “rules” of the process don’t change over time.

Counting Processes

Counting Process

A **counting process** $\{N(t), t \geq 0\}$ counts the number of events that have occurred by time t . It satisfies:

1. $N(t) \geq 0$ for all t
2. $N(t)$ is integer-valued
3. If $s < t$, then $N(s) \leq N(t)$ (non-decreasing)
4. $N(t) - N(s)$ = number of events in the interval $(s, t]$

Poisson Process (Lectures: Jan 6 & Jan 9)

What is a Poisson Process?

Poisson Process — Definition

A counting process $\{N(t), t \geq 0\}$ is a **Poisson process** with rate $\lambda > 0$ if:

1. $N(0) = 0$
2. It has **independent increments**
3. It has **stationary increments**
4. The number of events in any interval of length t follows a Poisson distribution:

$$P(N(t+s) - N(s) = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}, \quad k = 0, 1, 2, \dots$$

Intuitive Understanding

Imagine events happening “randomly” over time at an average rate of λ per unit time. A Poisson process is the simplest model for such random events:

- Events happen one at a time (no simultaneous events)
- The chance of an event in a tiny interval $[t, t + \Delta t)$ is $\approx \lambda \Delta t$
- What happened in the past doesn’t influence the future

Properties of the Poisson Distribution

Poisson Distribution Properties

If $X \sim \text{Poisson}(\lambda t)$:

$$P(X = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}$$

$$E[X] = \lambda t$$

$$\text{Var}(X) = \lambda t$$

Key property: Mean = Variance = λt .

Exponential Distribution — Inter-arrival Times

Inter-arrival Times are Exponential

If $\{N(t)\}$ is a Poisson process with rate λ , then the time between consecutive events (called **inter-arrival times**) $X_1, X_2, X_3, \dots$ are **i.i.d. Exponential(λ)** random variables:

$$P(X_i > t) = e^{-\lambda t}, \quad t \geq 0$$

Exponential Distribution Properties

If $X \sim \text{Exp}(\lambda)$:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$F(x) = 1 - e^{-\lambda x}$$

$$E[X] = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless Property

Memoryless Property of Exponential Distribution

The exponential distribution is the *only* continuous distribution that is **memoryless**:

$$P(X > s + t \mid X > s) = P(X > t)$$

Meaning: If you have already waited s time units, the remaining wait time has the *same* distribution as the original — the process “forgets” that time has passed.

Example: Light Bulb

A light bulb has an exponentially distributed lifetime with mean 1000 hours. If it has already been on for 500 hours, the probability it lasts another 500 hours is the *same* as the probability a brand-new bulb lasts 500 hours. It doesn’t “age.”

Merging (Superposition) of Poisson Processes

Merging of Poisson Processes

If $\{N_1(t)\}$ and $\{N_2(t)\}$ are *independent* Poisson processes with rates λ_1 and λ_2 , then their sum:

$$N(t) = N_1(t) + N_2(t)$$

is a Poisson process with rate $\lambda_1 + \lambda_2$.

Generalization: Merging n independent Poisson processes with rates $\lambda_1, \dots, \lambda_n$ gives a Poisson process with rate $\lambda_1 + \dots + \lambda_n$.

Splitting (Thinning) of Poisson Processes

Splitting of Poisson Processes

Given a Poisson process with rate λ , if each event is independently classified as “Type 1” with probability p and “Type 2” with probability $1 - p$, then:

- Type 1 events form a Poisson process with rate λp
- Type 2 events form a Poisson process with rate $\lambda(1 - p)$
- The two resulting processes are *independent*

Discrete-Time Markov Chains — DTMC (Lecture: Jan 13)

The Markov Property

Markov Property (Memoryless Property for Chains)

A stochastic process $\{X_n, n = 0, 1, 2, \dots\}$ is a **Markov chain** if:

$$P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j \mid X_n = i)$$

In plain English: The **future** depends on the **present** only, not on the **past**. The process has “no memory” of how it reached the current state.

Simple Example: Coin Toss Chain

Toss a fair coin repeatedly. Let X_n be the outcome of toss n (H or T). Since each toss is independent, the next outcome depends only on chance, not on previous outcomes. This satisfies the Markov property.

Transition Probability Matrix (TPM)

Transition Probability Matrix

For a Markov chain with states $\{0, 1, 2, \dots\}$, the **one-step transition probability** is:

$$p_{ij} = P(X_{n+1} = j \mid X_n = i)$$

The **Transition Probability Matrix (TPM)** P has entries p_{ij} :

$$P = \begin{pmatrix} p_{00} & p_{01} & p_{02} & \cdots \\ p_{10} & p_{11} & p_{12} & \cdots \\ p_{20} & p_{21} & p_{22} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Properties:

1. All entries are non-negative: $p_{ij} \geq 0$
2. Each row sums to 1: $\sum_j p_{ij} = 1$ for every i (it's a *stochastic matrix*)

Time Homogeneity

Time-Homogeneous Markov Chain

A Markov chain is **time-homogeneous** if the transition probabilities do *not* depend on the time step n :

$$P(X_{n+1} = j \mid X_n = i) = p_{ij} \quad \text{for all } n$$

Meaning: The “rules” for moving between states are the same at every time step.

Chapman–Kolmogorov Equation

Chapman–Kolmogorov Equation

The n -step transition probability $p_{ij}^{(n)} = P(X_n = j \mid X_0 = i)$ satisfies:

$$p_{ij}^{(n+m)} = \sum_k p_{ik}^{(n)} \cdot p_{kj}^{(m)}$$

In matrix form: $P^{(n+m)} = P^{(n)} \cdot P^{(m)}$, and in particular:

$$P^{(n)} = P^n \quad (n\text{-th power of the TPM})$$

Meaning: To go from state i to state j in $n + m$ steps, you pass through some intermediate state k after n steps.

Classification of States

Property	Definition
Accessible	State j is <i>accessible</i> from i if $p_{ij}^{(n)} > 0$ for some $n \geq 1$
Communicating	States i and j <i>communicate</i> if each is accessible from the other
Irreducible	The chain is <i>irreducible</i> if all states communicate with each other
Periodic	State i has <i>period</i> d if $d = \gcd\{n : p_{ii}^{(n)} > 0\}$
Aperiodic	State i is <i>aperiodic</i> if $d = 1$

Recurrence and Transience

Recurrent vs. Transient States

Let f_{ii} = probability of eventually returning to state i starting from i .

- State i is **recurrent** if $f_{ii} = 1$ (you *will* return with probability 1)
- State i is **transient** if $f_{ii} < 1$ (there is a positive probability of never returning)

Sub-classification of recurrent states:

- **Positive recurrent:** Expected return time $E[T_i] < \infty$ (finite mean return time)
- **Null recurrent:** $f_{ii} = 1$ but $E[T_i] = \infty$ (you will return, but on average it takes infinitely long)

Important Facts

For an irreducible Markov chain:

- *All* states are of the same type (all recurrent or all transient)
- If the state space is *finite*, the chain is always positive recurrent

- A finite irreducible aperiodic Markov chain always has a unique stationary distribution

Ergodicity

Ergodic Markov Chain

A Markov chain is **ergodic** if it is:

1. **Irreducible** — all states communicate
2. **Aperiodic** — $\gcd\{n : p_{ii}^{(n)} > 0\} = 1$ for all i
3. **Positive recurrent** — expected return time is finite

An ergodic chain has a *unique* stationary distribution, and the time average equals the ensemble average (more on this later).

Continuous-Time Markov Chains — CTMC (Lectures: Jan 16 & Jan 20)

From DTMC to CTMC

Why CTMC?

In a DTMC, transitions happen at discrete time steps $n = 0, 1, 2, \dots$. But in real systems (e.g., packet arrivals at a router), events can happen at *any* time. A **Continuous-Time Markov Chain (CTMC)** allows transitions at any point in continuous time.

CTMC — Definition

A stochastic process $\{X(t), t \geq 0\}$ taking values in a countable state space is a **CTMC** if:

$$P(X(t+s) = j \mid X(s) = i, \{X(u), 0 \leq u < s\}) = P(X(t+s) = j \mid X(s) = i)$$

Key idea: The Markov property holds in continuous time — the future depends only on the present state.

Sojourn (Holding) Times

Sojourn Times are Exponential

In a CTMC, the time spent in each state i before transitioning (called the **sojourn time** or **holding time**) is:

$$\text{Sojourn time in state } i \sim \text{Exp}(q_i)$$

where $q_i > 0$ is the total rate of leaving state i .

Why exponential? Because the exponential distribution is the *only* continuous memoryless distribution, and the Markov property requires memorylessness.

Transition Rate Matrix (Q-Matrix)

Q-Matrix (Generator Matrix)

The **Q-matrix** (or **generator matrix** or **rate matrix**) $Q = [q_{ij}]$ describes the transition rates:

$$Q = \begin{pmatrix} -q_0 & q_{01} & q_{02} & \cdots \\ q_{10} & -q_1 & q_{12} & \cdots \\ q_{20} & q_{21} & -q_2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Properties:

1. Off-diagonal entries $q_{ij} \geq 0$ for $i \neq j$ (transition rates)
2. Diagonal entries: $q_{ii} = -\sum_{j \neq i} q_{ij}$ (negative of total rate out)
3. Each row sums to zero: $\sum_j q_{ij} = 0$

Understanding the Q-Matrix

- q_{ij} ($i \neq j$): rate at which the process transitions from state i to state j
- $|q_{ii}| = q_i$: total rate of leaving state i
- Average time in state $i = 1/q_i$
- Probability of jumping to state j (given a jump from i) = q_{ij}/q_i

Transition Probability Matrix $P(t)$

Relationship Between $P(t)$ and Q

The transition probability matrix $P(t)$ satisfies:

$$\frac{dP(t)}{dt} = P(t) \cdot Q = Q \cdot P(t)$$

These are called the **Kolmogorov forward and backward equations**.

The solution is the **matrix exponential**:

$$P(t) = e^{Qt}$$

Chapman–Kolmogorov for CTMC:

$$P(t+s) = P(t) \cdot P(s)$$

How to Simulate a CTMC

Simulation Algorithm

To generate a sample path of a CTMC starting in state i :

1. Set current state = i , current time = 0
2. Sample the sojourn time: $T \sim \text{Exp}(q_i)$ where $q_i = |q_{ii}|$
3. Update time: $t \leftarrow t + T$
4. Choose next state j with probability q_{ij}/q_i (for $j \neq i$)
5. Set current state = j
6. Repeat from step 2

Alternative (competing exponentials):

- For each possible next state j , start an independent $\text{Exp}(q_{ij})$ clock

- The first clock to ring determines the next state and the sojourn time
- This works because min of independent exponentials is exponential with rate = sum of rates

Embedded DTMC

Embedded Discrete-Time Markov Chain

From a CTMC, we can extract an **embedded DTMC** by observing only the sequence of states visited (ignoring the times). The transition probabilities of the embedded chain are:

$$\tilde{p}_{ij} = \frac{q_{ij}}{q_i} \quad \text{for } j \neq i, \quad \tilde{p}_{ii} = 0$$

This is useful for analyzing the long-run behavior of the CTMC.

Stationary & Limiting Distributions (Lectures: Jan 20, 23, 30)

Stationary Distribution

Stationary Distribution

A probability distribution $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2, \dots)$ is a **stationary distribution** of a Markov chain if it does *not change* when multiplied by the transition mechanism:

For DTMC: $\boldsymbol{\pi} = \boldsymbol{\pi}P$ (i.e., $\pi_j = \sum_i \pi_i p_{ij}$ for all j)

For CTMC: $\boldsymbol{\pi}Q = \mathbf{0}$ (i.e., $\sum_i \pi_i q_{ij} = 0$ for all j)

with $\sum_i \pi_i = 1$ and $\pi_i \geq 0$ for all i .

Intuition: If you start the chain according to $\boldsymbol{\pi}$, the distribution *stays* $\boldsymbol{\pi}$ at all future times. It is the “equilibrium” of the system.

Limiting Distribution

Limiting Distribution

The **limiting distribution** is the probability distribution the chain converges to as time goes to infinity:

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \pi_j \quad (\text{DTMC}) \quad \lim_{t \rightarrow \infty} p_{ij}(t) = \pi_j \quad (\text{CTMC})$$

regardless of the starting state i .

For an **irreducible, aperiodic, positive recurrent** chain, the limiting distribution *exists*, is *unique*, and *equals* the stationary distribution.

Global Balance Equations

Global Balance

For a CTMC in steady state, the rate of probability “flow” *into* each state equals the rate of flow

out:

$$\underbrace{\sum_{i \neq j} \pi_i q_{ij}}_{\text{rate into state } j} = \underbrace{\pi_j q_j}_{\text{rate out of state } j}$$

where $q_j = \sum_{k \neq j} q_{jk}$ is the total rate out of state j .

This is equivalent to $\pi Q = \mathbf{0}$.

Detailed Balance Equations

Detailed Balance

A *stronger* condition than global balance. The chain satisfies **detailed balance** if for every pair of states i, j :

$$\pi_i q_{ij} = \pi_j q_{ji}$$

Meaning: The rate of transitions from i to j (in stationarity) equals the rate from j to i .

Why useful? If detailed balance holds, you can solve for π by writing pairwise equations — much simpler than solving the full global balance system.

A chain satisfying detailed balance is called **reversible**: a video of the chain played forward looks statistically identical to the video played backward.

When Does Detailed Balance Hold?

Detailed balance holds for:

- **Birth-death processes** (transitions only to neighboring states)
- Chains with a **tree-structured** state transition graph
- Many queueing systems (M/M/1, M/M/ ∞ , etc.)

It does *not* hold for all chains (e.g., cycles with asymmetric rates).

Ergodic Theorem — Time Average = Ensemble Average

Ergodic Theorem

For an ergodic Markov chain with stationary distribution π :

$$\text{Time average: } \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \mathbf{1}_{X(t)=j} dt = \pi_j \quad (\text{almost surely})$$

Meaning: The *fraction of time* spent in state j converges to π_j . You don't need many independent runs — one long run is enough.

More generally, for any function f :

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(X(t)) dt = \sum_j \pi_j f(j) = E_\pi[f(X)]$$

M/M/1 Queue (Lectures: Jan 30 & Feb 3)

Model Description

M/M/1 Queue

- **Arrivals:** Poisson process with rate λ (“M” for Markovian/Memoryless)
- **Service times:** i.i.d. Exponential with rate μ (“M” for Markovian)
- **Servers:** 1
- **Capacity:** ∞ (unlimited waiting room)
- **Discipline:** FCFS (First Come, First Served)

State: $X(t)$ = number of customers in the system at time t . State space = $\{0, 1, 2, \dots\}$.

Birth-Death Process Model

The M/M/1 queue is a **birth-death process**: from state n , the chain can only move to $n+1$ (arrival) or $n-1$ (service completion).

Transition	Rate
$n \rightarrow n+1$ (arrival)	λ for all $n \geq 0$
$n \rightarrow n-1$ (departure)	μ for all $n \geq 1$

Q-Matrix for M/M/1

Q-Matrix

$$Q = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & \cdots \\ \mu & -(\lambda + \mu) & \lambda & 0 & \cdots \\ 0 & \mu & -(\lambda + \mu) & \lambda & \cdots \\ \vdots & & \ddots & \ddots & \ddots \end{pmatrix}$$

Stationary Distribution of M/M/1

M/M/1 Stationary Distribution

Using detailed balance: $\pi_n \lambda = \pi_{n+1} \mu$, which gives $\pi_{n+1} = \rho \pi_n$ where $\rho = \lambda/\mu$.

Solving with $\sum_{n=0}^{\infty} \pi_n = 1$:

$$\pi_n = (1 - \rho) \rho^n, \quad n = 0, 1, 2, \dots$$

This is a **geometric distribution**.

Condition: $\rho < 1$ (system must be stable).

$\pi_0 = 1 - \rho$ is the probability the system is *empty* (server is idle).

Performance Metrics for M/M/1

M/M/1 Performance Formulas

Metric	Formula	Meaning
Utilization	$\rho = \frac{\lambda}{\mu}$	Fraction of time server is busy
Avg. # in system	$L = \frac{\rho}{1 - \rho}$	Expected number of customers
Avg. # in queue	$L_q = \frac{\rho^2}{1 - \rho}$	Expected number waiting
Avg. time in system	$W = \frac{1}{\mu - \lambda}$	Expected total time
Avg. wait in queue	$W_q = \frac{\rho}{\mu - \lambda}$	Expected waiting time

Worked Example

A web server receives requests at rate $\lambda = 8$ per second and can process them at rate $\mu = 10$ per second.

$$\rho = 8/10 = 0.8$$

$$L = 0.8/(1 - 0.8) = 0.8/0.2 = 4 \text{ requests on average}$$

$$W = 1/(10 - 8) = 0.5 \text{ seconds}$$

$$W_q = 0.8/(10 - 8) = 0.4 \text{ seconds waiting}$$

If we upgrade the server to $\mu = 12$: $\rho = 0.667$, $L = 2$, $W = 0.25$ s — halved!

M/M/ ∞ Queue (Lecture: Feb 3)**M/M/ ∞ Queue**

- **Arrivals:** Poisson process with rate λ
- **Service times:** i.i.d. Exponential with rate μ
- **Servers:** ∞ (every customer gets their own server immediately — no waiting!)

This models systems with abundant resources (e.g., a large cloud with auto-scaling).

Q-Matrix and Transition Rates

Transition	Rate
$n \rightarrow n + 1$ (arrival)	λ
$n \rightarrow n - 1$ (departure)	$n\mu$ (because n servers are busy)

Stationary Distribution**M/M/ ∞ Stationary Distribution**

Using detailed balance: $\pi_n \lambda = \pi_{n+1} (n+1) \mu$, which gives:

$$\pi_{n+1} = \frac{\lambda}{(n+1)\mu} \pi_n = \frac{\rho}{n+1} \pi_n$$

Solving recursively with $\rho = \lambda/\mu$:

$$\pi_n = e^{-\rho} \frac{\rho^n}{n!}, \quad n = 0, 1, 2, \dots$$

This is a **Poisson distribution** with parameter $\rho = \lambda/\mu$.

Note: This queue is *always stable* (no condition on ρ) because adding servers prevents buildup.

M/M/ ∞ Performance

$$L = \rho = \frac{\lambda}{\mu} \quad (\text{average number in system})$$

$$W = \frac{1}{\mu} \quad (\text{average time in system} = \text{just service time, no waiting!})$$

Tandem Queues & Product Form Solutions (Lecture: Feb 3)

Tandem Queue

Tandem Queue

A **tandem queue** consists of two (or more) queues *in series*: a customer completes service at Queue 1, then immediately joins Queue 2.

Setup:

- External arrivals to Queue 1: Poisson with rate λ
- Queue 1: single server with rate μ_1
- Queue 2: single server with rate μ_2
- State: (n_1, n_2) where n_i = number of customers at Queue i

Burke's Theorem

Burke's Theorem

The **departure process** from a stationary M/M/1 queue with arrival rate λ is also a **Poisson process with rate λ** .

Implication: In a tandem of M/M/1 queues, the arrivals to Queue 2 are Poisson $\rightarrow$ Queue 2 is also M/M/1.

Product Form Solution

Product Form for Tandem Queue

For a tandem of two M/M/1 queues, the stationary distribution has a **product form**:

$$\pi(n_1, n_2) = \pi_1(n_1) \cdot \pi_2(n_2)$$

where $\pi_k(n_k) = (1 - \rho_k)\rho_k^{n_k}$ with $\rho_k = \lambda/\mu_k$.

Meaning: In steady state, the two queues behave *independently*! You can analyze each queue separately.

Little's Law (Lectures: Feb 3, Feb 24)

Little's Law

For *any* stable queueing system (not just Markovian!):

$$L = \lambda W$$

where:

- L = average number of customers in the system
- λ = average arrival rate
- W = average time a customer spends in the system

Why Little's Law is Powerful

- It is *completely general* — works for any queue discipline, any arrival process, any service distribution
- It doesn't require Markov assumptions
- You can apply it to subsystems: e.g., $L_q = \lambda W_q$ (queue only) or $L_s = \lambda W_s$ (server only)
- If you know any two of $\{L, \lambda, W\}$, you can find the third

Applying Little's Law to M/M/1

For M/M/1: $W = 1/(\mu - \lambda)$.

$$L = \lambda \cdot W = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho} \quad \checkmark$$

For the queue subsystem: $W_q = W - 1/\mu = \rho/(\mu - \lambda)$.

$$L_q = \lambda \cdot W_q = \frac{\lambda \rho}{\mu - \lambda} = \frac{\rho^2}{1 - \rho} \quad \checkmark$$

Renewal Theory (Lectures: Feb 10 & Feb 17)

Renewal Processes

Renewal Process

A **renewal process** is a counting process where the inter-arrival times $X_1, X_2, X_3, \dots$ are **i.i.d. non-negative** random variables (with any distribution, not just exponential).

Let $S_n = X_1 + X_2 + \dots + X_n$ be the time of the n -th event. Define:

$$N(t) = \max\{n : S_n \leq t\} = \text{number of renewals by time } t$$

Special case: If $X_i \sim \text{Exp}(\lambda)$, the renewal process is a Poisson process.

Example: Machine Replacement

A machine fails after a random time. Each time it fails, it is replaced with a new (identical) machine. The times between replacements are i.i.d. with distribution F . The number of replacements by time t is a renewal process.

Renewal Equation

Renewal Equation and Its Solution

Let $m(t) = E[N(t)]$ be the expected number of renewals by time t , and $F(t) = P(X_1 \leq t)$. Then:

$$m(t) = F(t) + \int_0^t m(t-x) f(x) dx$$

where $f(x)$ is the density of the inter-arrival time.

Taking **Laplace transforms** ($\tilde{g}(s) = \int_0^\infty e^{-st} g(t) dt$):

$$\tilde{m}(s) = \frac{\tilde{F}(s)}{1 - \tilde{f}(s)}$$

This makes the integral equation into a simple algebraic equation!

Convolution

Convolution

The **convolution** of two functions f and g is:

$$(f * g)(t) = \int_0^t f(x) g(t-x) dx$$

Meaning: If X and Y are independent with densities f and g , then $X + Y$ has density $f * g$.

Key property: The Laplace transform turns convolution into multiplication:

$$\widetilde{f * g}(s) = \tilde{f}(s) \cdot \tilde{g}(s)$$

Laplace Transform

Laplace Transform

The Laplace transform of a function $f(t)$ is:

$$\tilde{f}(s) = \mathcal{L}\{f\}(s) = \int_0^\infty e^{-st} f(t) dt$$

Common transforms:

$f(t)$	$\tilde{f}(s)$
$e^{-\lambda t}$	$\frac{1}{s + \lambda}$
$\lambda e^{-\lambda t}$ (Exponential density)	$\frac{\lambda}{s + \lambda}$
t^n	$\frac{n!}{s^{n+1}}$
1	$\frac{1}{s}$

Key property: $\mathcal{L}\{f * g\} = \tilde{f} \cdot \tilde{g}$ (convolution becomes multiplication).

Moment Generating Function (MGF)

Moment Generating Function

The **MGF** of a random variable X is:

$$M_X(t) = E[e^{tX}]$$

Why useful?

- $M'_X(0) = E[X]$ (first moment)
- $M''_X(0) = E[X^2]$ (second moment)
- In general: $M^{(n)}_X(0) = E[X^n]$ (n -th moment)

Connection to Laplace: For a non-negative r.v. X with density f :

$$\tilde{f}(s) = E[e^{-sX}] = M_X(-s)$$

Elementary Renewal Theorem

Elementary Renewal Theorem

For a renewal process with inter-arrival times having mean $E[X] = \mu_X$:

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu_X} \quad (\text{almost surely})$$

and

$$\lim_{t \rightarrow \infty} \frac{E[N(t)]}{t} = \frac{1}{\mu_X}$$

Meaning: In the long run, renewals occur at rate $1/\mu_X$ (one renewal every μ_X time units on average).

Renewal Reward Theorem (Lectures: Feb 17, Feb 24)

Renewal Reward Theorem

Suppose rewards $R_1, R_2, \dots$ are earned in successive renewal cycles with durations $X_1, X_2, \dots$. Then the long-run average reward rate is:

$$\lim_{t \rightarrow \infty} \frac{\text{Total reward in } [0, t]}{t} = \frac{E[R]}{E[X]}$$

Meaning: Long-run rate = expected reward per cycle / expected cycle length.

Application: Server Utilization

In an M/G/1 queue, a “cycle” is a busy period + idle period.

- Reward = time the server is busy in one cycle
- Expected cycle = $1/\lambda + E[B]/(1 - \rho)$ (idle + busy)
- Server utilization = $E[\text{busy time}]/E[\text{cycle}] = \rho$

Stopping Times & Strong Markov Property (Lecture: Feb 17)

Stopping Time

A random variable τ is a **stopping time** with respect to a process $\{X_n\}$ if the event $\{\tau = n\}$ depends *only* on $X_0, X_1, \dots, X_n$ (the present and past), *not* on the future.

Intuition: A stopping time is a time at which you can decide to “stop” based only on what you’ve seen so far.

Examples (valid stopping times):

- First time a random walk hits level +5
- First time the number of customers in a queue exceeds 10

Non-examples:

- The last time the walk visits 0 before time T (requires future knowledge)

Strong Markov Property

If $\{X_n\}$ is a Markov chain and τ is a stopping time, then $\{X_{\tau+n}\}$ is also a Markov chain with the same transition probabilities, starting from X_τ .

Meaning: The Markov property holds not just at fixed times but also at random stopping times.

M/G/1 Queue, Residual Time & Inspection Paradox (Lecture: Feb 24)

M/G/1 Queue

M/G/1 Queue

- **Arrivals:** Poisson with rate λ
- **Service times:** i.i.d. with *general* distribution G (mean $b = E[S]$, second moment $b_2 = E[S^2]$)
- **Servers:** 1

The service time need *not* be exponential — it can be *any* distribution.

Residual Time (Excess Life)

Residual Time

Suppose events occur according to a renewal process with inter-arrival distribution F . If you arrive at a “random” time t , the **residual time** $R(t)$ is the time *remaining* until the next event.

The long-run (limiting) expected residual time is:

$$E[R] = \frac{E[X^2]}{2 E[X]}$$

where X is the inter-arrival time.

Inspection Paradox

The Inspection Paradox

You might expect the residual time to be $E[X]/2$. But actually:

$$E[R] = \frac{E[X^2]}{2 E[X]} = \frac{E[X]}{2} + \frac{\text{Var}(X)}{2 E[X]} \geq \frac{E[X]}{2}$$

The residual time is *always at least half the mean*, with equality only when X is deterministic.

Why? When you arrive at a random time, you are more likely to land in a *long* interval than a short one. This **length bias** inflates the expected residual time.

Bus Waiting Paradox

Buses arrive with inter-arrival times having mean 10 min and standard deviation 5 min.

$$E[X] = 10, \quad E[X^2] = \text{Var}(X) + (E[X])^2 = 25 + 100 = 125$$

$$E[R] = \frac{125}{2 \times 10} = 6.25 \text{ min}$$

You’d naively expect to wait 5 min ($= 10/2$), but the actual average wait is 6.25 min!

Pollaczek–Khinchine (P-K) Mean Value Formula

P-K Formula for M/G/1 Average Waiting Time

For an M/G/1 queue with arrival rate λ , mean service time b , and second moment of service time $b_2 = E[S^2]$:

$$W_q = \frac{\lambda b_2}{2(1 - \rho)}$$

where $\rho = \lambda b$.

The total time in the system:

$$W = W_q + b = \frac{\lambda b_2}{2(1 - \rho)} + b$$

Using Little’s Law: $L = \lambda W$, $L_q = \lambda W_q$.

Understanding the P-K Formula

The average waiting time W_q has a clean interpretation:

$$W_q = \underbrace{\frac{\lambda b_2}{2}}_{\text{avg. residual service time of current job}} \cdot \underbrace{\frac{1}{1-\rho}}_{\text{amplification due to queued jobs}}$$

- The first factor is the expected remaining work the server is doing when you arrive
- The second factor accounts for the work from jobs already waiting ahead of you
- Higher variance in service time $\Rightarrow$ higher $b_2 \Rightarrow$ longer waits!

M/M/1 as Special Case of M/G/1

For exponential service times: $b = 1/\mu$, $b_2 = 2/\mu^2$:

$$W_q = \frac{\lambda \cdot 2/\mu^2}{2(1 - \lambda/\mu)} = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{\rho}{\mu - \lambda}$$

This matches our M/M/1 formula. ✓

Introduction to Reinforcement Learning (Lecture: Feb 20)

Reinforcement Learning (RL)

Reinforcement Learning is a type of machine learning where an *agent* learns to make *sequential decisions* by interacting with an *environment* and receiving *rewards* (or penalties).

Unlike supervised learning (where you have labeled data), in RL the agent learns through **trial and error**.

Key Concepts

Concept	Explanation
Agent	The decision-maker (e.g., a robot, a game-playing AI)
Environment	The world the agent interacts with
State	Current situation/configuration
Action	What the agent can do in a given state
Reward	Feedback signal (positive or negative) after taking an action
Policy	A strategy mapping states to actions
Value Function	Expected cumulative reward from a state under a policy

Static vs. Dynamic Optimization

	Static Optimization	Dynamic Optimization (RL)
Time	One-shot decision	Sequence of decisions over time
Information	All info available upfront	Info revealed gradually
Example	Find min of $f(x)$	Play chess, control a robot

Connection to Markov Decision Processes (MDPs)

RL and MDPs

RL problems are typically modeled as **Markov Decision Processes (MDPs)**:

- States, actions, transition probabilities, and rewards
- The Markov property: the next state depends only on the current state and action
- Goal: find the policy that maximizes the expected cumulative reward

This connects directly to the Markov chains we studied — MDPs add the concept of *actions* (choices) at each state.

Games as Testbeds

Games (chess, Go, Atari) are popular RL testbeds because:

- Well-defined rules and states
- Clear objective (win/score)
- Challenging enough to require sophisticated strategies
- Famous example: AlphaGo (DeepMind) used RL + neural networks to beat human champions

Quick Reference — All Key Formulas

Exponential Distribution $X \sim \text{Exp}(\lambda)$

$$f(x) = \lambda e^{-\lambda x}, \quad E[X] = 1/\lambda, \quad \text{Var}(X) = 1/\lambda^2$$

Memoryless: $P(X > s + t \mid X > s) = P(X > t)$

Poisson Distribution $X \sim \text{Poisson}(\mu)$

$$P(X = k) = \frac{e^{-\mu} \mu^k}{k!}, \quad E[X] = \mu, \quad \text{Var}(X) = \mu$$

DTMC Stationary Distribution

Solve $\pi = \pi P$ with $\sum_i \pi_i = 1$.

CTMC Stationary Distribution

Solve $\pi Q = \mathbf{0}$ with $\sum_i \pi_i = 1$.

Detailed Balance (CTMC)

$\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j .

M/M/1 Queue

$$\pi_n = (1 - \rho)\rho^n, \quad L = \frac{\rho}{1 - \rho}, \quad W = \frac{1}{\mu - \lambda}$$

$$L_q = \frac{\rho^2}{1 - \rho}, \quad W_q = \frac{\rho}{\mu - \lambda}, \quad \rho = \lambda/\mu < 1$$

M/M/ ∞ Queue

$$\pi_n = e^{-\rho} \frac{\rho^n}{n!}, \quad L = \rho, \quad W = 1/\mu$$

Little's Law

$$L = \lambda W, \quad L_q = \lambda W_q, \quad L_s = \lambda W_s$$

Renewal Equation (Laplace domain)

$$\tilde{m}(s) = \frac{\tilde{F}(s)}{1 - \tilde{f}(s)}$$

Renewal Reward Theorem

$$\text{Long-run rate} = \frac{E[\text{reward per cycle}]}{E[\text{cycle length}]}$$

Expected Residual Time

$$E[R] = \frac{E[X^2]}{2 E[X]}$$

P-K Formula (M/G/1)

$$W_q = \frac{\lambda E[S^2]}{2(1 - \rho)}, \quad W = W_q + E[S], \quad \rho = \lambda E[S]$$

Chapman–Kolmogorov

$$\text{DTMC: } P^{(n+m)} = P^{(n)} P^{(m)} \quad \text{CTMC: } P(t+s) = P(t)P(s), \quad P(t) = e^{Qt}$$

Important Concepts — Quick Cheat Sheet

Concept	One-Line Summary
Stochastic Process	Collection of random variables indexed by time
Counting Process	Counts number of events by time t
Poisson Process	Counting process with Exp inter-arrivals; ind. & stat. increments
Exponential Dist.	Only continuous memoryless distribution
DTMC	Chain with discrete time; future depends only on present
CTMC	Chain with continuous time; sojourn times are exponential
TPM (P)	Matrix of one-step transition probabilities (rows sum to 1)
Q-Matrix	Matrix of transition rates (rows sum to 0)
Stationary Dist.	Distribution unchanged by transitions: $\pi P = \pi$ or $\pi Q = 0$
Limiting Dist.	Distribution the chain converges to as $t \rightarrow \infty$
Global Balance	Rate in = Rate out for each state
Detailed Balance	Pairwise rate balance: $\pi_i q_{ij} = \pi_j q_{ji}$
Irreducible	All states communicate
Aperiodic	GCD of return times is 1
Recurrent	Probability of return = 1
Positive Recurrent	Expected return time is finite
Ergodic	Irreducible + aperiodic + positive recurrent
Little's Law	$L = \lambda W$ (universal)
Renewal Process	Events with i.i.d. inter-arrival times
Renewal Equation	$m(t) = F(t) + \int_0^t m(t-x)f(x)dx$
Laplace Transform	$\tilde{f}(s) = \int_0^\infty e^{-st}f(t)dt$; turns convolution into multiplication
MGF	$M_X(t) = E[e^{tX}]$; derivatives give moments
Stopping Time	Random time depending only on past and present
Residual Time	Remaining time until next renewal; $E[R] = E[X^2]/(2E[X])$
Inspection Paradox	You land in longer intervals more often
P-K Formula	$W_q = \lambda E[S^2]/(2(1-\rho))$ for M/G/1
Burke's Theorem	Departures from M/M/1 in steady state are Poisson
Product Form	Tandem queues decompose: $\pi(n_1, n_2) = \pi_1(n_1)\pi_2(n_2)$

Practice Exam Questions with Solutions

Q1: Poisson Process

Q: Customers arrive at a shop according to a Poisson process with rate $\lambda = 5$ per hour. What is the probability that exactly 3 customers arrive in 30 minutes?

Solution: In 30 min = 0.5 hr, the number of arrivals $\sim \text{Poisson}(\lambda t) = \text{Poisson}(2.5)$:

$$P(N = 3) = \frac{e^{-2.5}(2.5)^3}{3!} = \frac{e^{-2.5} \times 15.625}{6} \approx 0.2138$$

Q2: M/M/1 Queue

Q: In an M/M/1 queue with $\lambda = 4$ and $\mu = 6$, find: (a) utilization, (b) average number in system, (c) average time in system, (d) probability system is empty.

Solution:

- (a) $\rho = 4/6 = 2/3$
- (b) $L = \rho/(1 - \rho) = (2/3)/(1/3) = 2$
- (c) $W = 1/(\mu - \lambda) = 1/(6 - 4) = 0.5$
- (d) $\pi_0 = 1 - \rho = 1/3$

Verify Little's Law: $L = \lambda W = 4 \times 0.5 = 2 \checkmark$

Q3: Detailed Balance

Q: For a birth-death process with birth rates $\lambda_n = \lambda$ and death rates $\mu_n = \mu$ (for $n \geq 1$), find the stationary distribution using detailed balance.

Solution: Detailed balance: $\pi_n \lambda = \pi_{n+1} \mu \Rightarrow \pi_{n+1} = (\lambda/\mu) \pi_n = \rho \pi_n$.

Recursively: $\pi_n = \rho^n \pi_0$.

Normalization: $\sum_{n=0}^{\infty} \rho^n \pi_0 = 1 \Rightarrow \pi_0 = 1 - \rho$ (for $\rho < 1$).

$$\pi_n = (1 - \rho) \rho^n$$

Q4: Residual Time

Q: Inter-arrival times have mean $E[X] = 20$ min and $E[X^2] = 500 \text{ min}^2$. Find the expected residual time and verify the inspection paradox.

Solution:

$$E[R] = \frac{E[X^2]}{2E[X]} = \frac{500}{2 \times 20} = 12.5 \text{ min}$$

Note: $E[X]/2 = 10$ min, but $E[R] = 12.5 > 10$. The extra 2.5 min is due to the inspection paradox (length bias).

Q5: M/G/1 P-K Formula

Q: An M/G/1 queue has $\lambda = 3$ per min, mean service time $b = 0.2$ min, and $E[S^2] = 0.06 \text{ min}^2$. Find the average waiting time in queue and average number in queue.

Solution:

$$\begin{aligned}\rho &= \lambda b = 3 \times 0.2 = 0.6 \\ W_q &= \frac{\lambda E[S^2]}{2(1-\rho)} = \frac{3 \times 0.06}{2(1-0.6)} = \frac{0.18}{0.8} = 0.225 \text{ min} \\ L_q &= \lambda W_q = 3 \times 0.225 = 0.675\end{aligned}$$

Q6: Renewal Reward Theorem

Q: A machine earns \$100 per hour while working. It works for $\text{Exp}(1/8)$ hours (mean 8 hrs), then breaks down and takes $\text{Exp}(1/2)$ hours (mean 2 hrs) to repair. What is the long-run earning rate?

Solution: One cycle = work time + repair time.

$$\begin{aligned}E[\text{cycle}] &= 8 + 2 = 10 \text{ hrs} \\ E[\text{reward per cycle}] &= 100 \times 8 = \$800 \\ \text{Long-run rate} &= \frac{800}{10} = \$80/\text{hr}\end{aligned}$$

Q7: Merging Poisson Processes

Q: Two types of requests arrive at a server: Type A at rate 3/min and Type B at rate 7/min, both Poisson. What is the distribution of the combined arrival process?

Solution: The merged process is Poisson with rate $3 + 7 = 10/\text{min}$.
Given an arrival occurs, the probability it is Type A is $3/10 = 0.3$.

Q8: Stationary Distribution of CTMC

Q: A CTMC has states $\{0, 1, 2\}$ with Q-matrix:

$$Q = \begin{pmatrix} -2 & 1 & 1 \\ 3 & -5 & 2 \\ 4 & 1 & -5 \end{pmatrix}$$

Find the stationary distribution.

Solution: Solve $\pi Q = \mathbf{0}$ with $\pi_0 + \pi_1 + \pi_2 = 1$:

$$\begin{aligned}-2\pi_0 + 3\pi_1 + 4\pi_2 &= 0 \\ \pi_0 - 5\pi_1 + \pi_2 &= 0 \\ \pi_0 + \pi_1 + \pi_2 &= 1\end{aligned}$$

From equation 2: $\pi_0 = 5\pi_1 - \pi_2$.

Substitute into equation 1: $-2(5\pi_1 - \pi_2) + 3\pi_1 + 4\pi_2 = 0 \Rightarrow -7\pi_1 + 6\pi_2 = 0 \Rightarrow \pi_2 = 7\pi_1/6$.

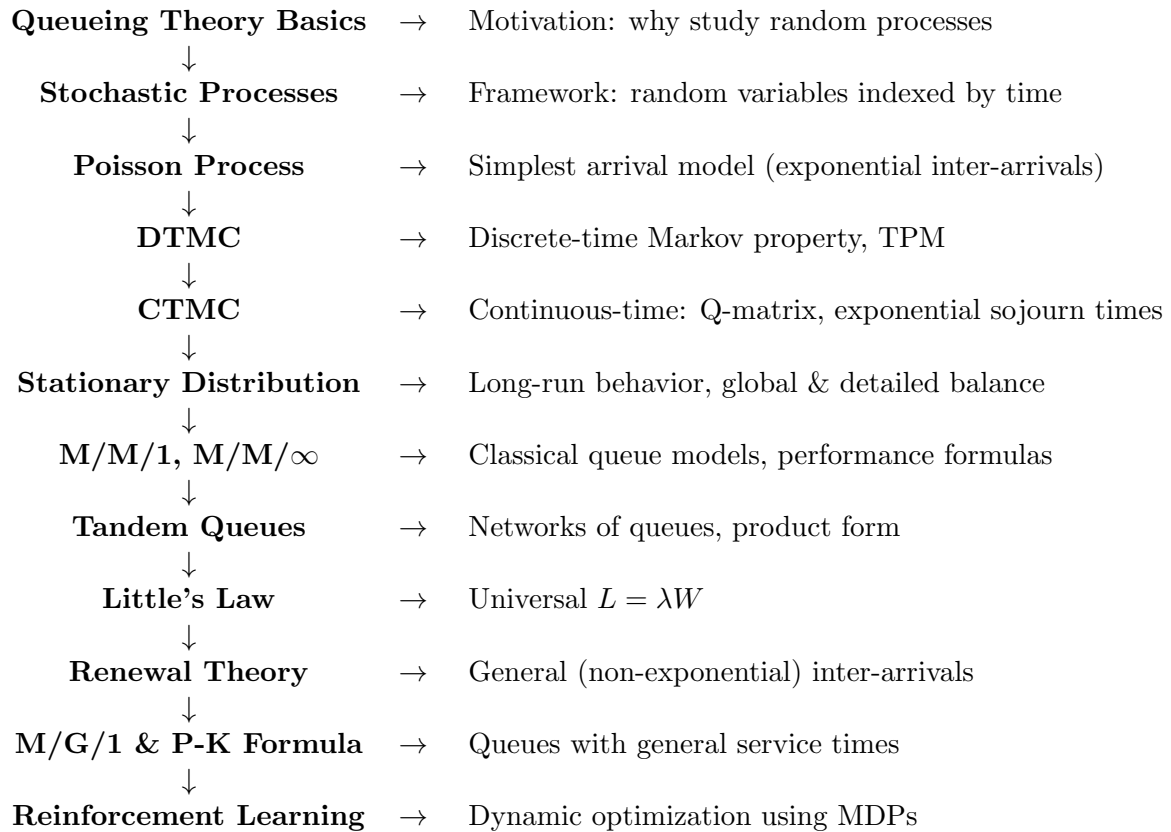
From normalization: $(5\pi_1 - 7\pi_1/6) + \pi_1 + 7\pi_1/6 = 1 \Rightarrow 6\pi_1 = 1 \Rightarrow \pi_1 = 1/6$.

So $\pi_2 = 7/36$, $\pi_0 = 5/6 - 7/36 = 23/36$.

$\pi = (23/36, 1/6, 7/36) \approx (0.639, 0.167, 0.194)$.

How Topics Connect — Course Flow

Course Topic Flow



Good luck on your exam!